

MCN v1.0

The modernclassnotes Class & Package

User Manual & Architecture Guide for Class Notes Package

Professor: Modern Class Notes Maintainers

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Open Source TeX Initiative

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1 Introduction

The `modernclassnotes` package and document class provides a modern, highly aesthetic, and open-source LaTeX framework designed specifically for university professors, lecturers, and educators. It enables effortless creation and distribution of lecture notes, student handouts, and problem worksheets.

[KEY] Key Takeaways

Key highlights of `modernclassnotes`:

- **Four Specialized Modes:** `notes` (full course manuals), `handout` (1–2 page lecture summaries), `worksheet` (homeworks and solution keys), and `chapternotes` (chapter-based notes and reading summaries).
- **Curated Color Palettes:** Switch between 8 themes (`ocean`, `nord`, `midnight`, `emerald`, `burgundy`, `amethyst`, `purple`, `mono`) with a single class option.
- **Zero Mandatory Heavy Dependencies:** Native TeX/LaTeX box engine ensuring maximum portability across TeX Live, MiKTeX, and Overleaf.
- **Lecture & Chapter Tracking:** Automated lecture header banners and chapter headers with optional objectives via `\lecture`, `\chapter`, and `\subchapter`.

2 Class Options & Configuration

Load the class in your LaTeX document header:

Loading the Class

```
1 \documentclass[mode=notes, palette=ocean, fontsize=11pt, font=default]{modernclassnotes}
```

2.1 Available Class Keys

Key	Values
<code>mode</code>	<code>notes</code> , <code>handout</code> , <code>worksheet</code> , <code>chapternotes</code>
<code>palette</code>	<code>ocean</code> , <code>midnight</code> , <code>nord</code> , <code>emerald</code> , <code>burgundy</code> , <code>amethyst</code> , <code>purple</code> , <code>mono</code>
<code>fontsize</code>	<code>10pt</code> , <code>11pt</code> , <code>12pt</code>
<code>font</code>	<code>default</code> , <code>palatino</code> , <code>charter</code> , <code>sans/helvet</code> , <code>kpfonts</code> , <code>pagella</code> , <code>sourceserif</code> , <code>fira</code> , <code>libe</code>
<code>official</code>	<code>true</code> , <code>false</code>
<code>showsolutions</code>	<code>true</code> , <code>false</code>
<code>parskip</code>	<code>true</code> , <code>false</code>

Table 1: Class options and parameter values

3 Course Metadata Commands

Specify your course metadata in the document preamble:

Metadata Setup

```
1 \course{Data Structures & Algorithms}
2 \coursecode{CS 201}
```

```

3 \professor{Dr. Alex Morgan}
4 \institution{Department of Computer Science}
5 \term{Fall 2026}
6 \docsubtitle{Comprehensive Lecture Notes}
7 \title{CS 201: Class Notes}

```

4 Lecture & Chapter Tracking Systems

Use the `\lecture` command to divide course notes into distinct lecture sessions:

Lecture Macro

```

1 \lecture[September 2, 2026]{1}{Introduction to Algorithm Analysis}

```

For chapter-based notes (`mode=chaptersnotes`), use `\chapter` and `\subchapter`:

Chapter Macro with Objectives

```

1 \chapter{1}{Digital Systems and Binary Numbers}{%
2   Understand binary numbers.,%
3   {Convert between binary, octal, decimal, and hexadecimal.},%
4   Form complements and parity bits.%
5 }
6
7 \subchapter{2}{Binary Numbers}

```

5 Pedagogical Environments

[DEF] Definition 1 (Mathematical Callout Environments)

`modernclassnotes` provides structured environments for mathematical and academic content:

- `theorem`, `lemma`, `proposition`, `corollary`, `proof`
- `definition`, `example`, `remark`, `warning`, `tip`, `takeaways`
- `exercise`, `solution`, `codebox`

[EX] Example 1 (Sample Theorem Environment)

```

1 \begin{theorem}[Euler's Identity]
2 For any real number  $\theta$ ,  $e^{i\theta} = \cos\theta + i\sin\theta$ .
3 \end{theorem}

```

6 Open-Source & CTAN Distribution Guidelines

To distribute this package under an open-source initiative:

1. Maintain the LPPL 1.3c / MIT dual licensing terms in `LICENSE`.
2. Provide compiled sample PDFs in `examples/` for GitHub previews.

3. Execute `make ctan` to bundle a clean `.zip` release file ready for CTAN submission.